

PRATHAM RAJ

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[Portfolio](#) | [LeetCode](#) | [Codeforces](#)

EDUCATION

National Institute of Technology Patna

B.Tech in Electrical Engineering; CGPA: 7.48 / 10.0

Patna, India

Aug 2023 - May 2027

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, TypeScript, HTML5, CSS3

Backend & Systems: Node.js, Express.js, RESTful APIs, Socket/Network Programming

Databases & Cloud: MongoDB, MySQL, Vercel, Cloudinary, Resend

Tools & Libraries: React.js, Tailwind CSS, Git/GitHub, Postman, Linux/Unix, CMake

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks, System Design, Multi-threading, Deep Packet Inspection

EXPERIENCE

Web Developer Intern

May 2026 - July 2026

[InternPe \(Certificate\)](#) | [GitHub](#) | [Live](#)

Remote

- Engineered an Interactive Question Sheet using React.js, establishing state management to process real-time user responses.
- Designed responsive UI components with Tailwind CSS, ensuring cross-device compatibility and increasing mobile user engagement.
- Deployed scalable frontend architecture via Vercel, optimizing asset delivery to achieve sub-second page load times.

PROJECTS

C++ Vector Database & RAG Engine | C++, [cpp-http-lib](#), [HNSW](#) | [GitHub](#)

Jan 2026 - Present

- Architected a multi-threaded C++ Vector Database, implementing HNSW and KD-Tree algorithms to achieve $O(\log N)$ search complexity across 10,000+ high-dimensional embeddings.
- Designed a local RAG pipeline utilizing Ollama and Llama 3.2 to embed and retrieve context, generating AI responses in under 2 seconds.
- Exposed a REST API via [cpp-http-lib](#) capable of handling 500+ concurrent CRUD operations, featuring real-time PCA scatter-plot visualizations.

NIT Patna Market | [React.js](#), [Node.js](#), [MongoDB](#) | [GitHub](#) | [Live](#)

Aug 2025 - Present

- Led the full-stack development of a campus marketplace using the MERN stack, establishing JWT authentication and Role-Based Access Control to securely serve 1,000+ university students.
- Automated secure OTP email verification using the Resend API and integrated Cloudinary CDN, reducing average image load times by 30%.
- Implemented real-time chat via HTTP polling and progressive web app (PWA) caching, enabling an optimistic UI capable of processing 500+ daily transactions.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

Amazon HackOn 6.0: Listed among the **Top 75 coders** out of 70,000+ participants globally.

Competitive Programming: Attained **Knight Badge** (2000+ peak rating) on LeetCode, **3-Star** (1600+) on CodeChef, and **Pupil** (1200+) on Codeforces.

Hackathon Champion: Secured **1st place** at the Hackslash College Hackathon, demonstrating rapid prototyping and algorithmic optimization under strict time constraints.

Industry Recognition: Shortlisted for the highly competitive Digital Specialist Engineer (DSE) role at Infosys based on top-percentile technical assessments.

Institutional Excellence: Ranked in the **Top 15** competitive programmers institution-wide at NIT Patna on the Codolio leaderboard.

LEADERSHIP & VOLUNTEERING

Campus Ambassador, HackerRank & GfG: Facilitated Data Structures & Algorithms (DSA) workshops, mentoring peers in competitive programming.

Educational Volunteer, NSS: Instructed 500+ students, spearheading community-driven educational outreach initiatives.

Technical Contributor, GDSC: Architected open-source software solutions, collaborating to foster campus developer communities.